
Autonomous Phenotyping for Agentic EHR-Based Clinical Research Workflows: Midterm Report

Irith Katiyar*

Department of Biomedical Informatics
Harvard Medical School
Boston, MA, USA
irith_katiyar@hms.harvard.edu

Benet Fité Abril

Department of Biomedical Informatics
Harvard Medical School
Boston, MA, USA
benet_fite@hms.harvard.edu

Abstract

Constructing reliable research cohorts from electronic health records remains a major bottleneck due to billing bias and miscoding in standard diagnostic code filtering. Our goal is to build an LLM-based agentic system that automates the full cohort phenotyping pipeline end-to-end from a single natural-language research prompt. The baseline model is an LLM agent that builds cohorts via SQL-based ICD filtering on MIMIC-IV, which addresses the natural-language interface but not the underlying phenotyping limitations. We extend it with vocabulary standardization, NLP feature extraction, and probabilistic phenotyping with MAP (multimodal automated phenotyping). As a case study on rheumatoid arthritis, our agent pipeline filters out about 60% of patients from the baseline cohort, consistent with prior studies showing substantial false positives from ICD codes alone. Our next steps include integrating LATTE for incident phenotyping from longitudinal data, running multi-condition ablation experiments to better quantify the contributions of each layer of our pipeline, and exploring LLM-as-a-judge evaluation.

1 Introduction and Project Overview

Electronic health records (EHR) represent the most comprehensive source of real-world clinical evidence available today, yet transforming them into reliable research cohorts remains a significant bottleneck for clinical research studies. A well-characterized pipeline for this transformation is complex and tedious [1], and the usual shortcut of filtering patients by combinations of diagnostic codes is known to suffer from billing bias, undercoding, and miscoding. For example, one study at MGB found that only 58% of patients bearing two or more ICD codes for rheumatoid arthritis were confirmed cases upon manual chart review [2]. More sophisticated phenotyping algorithms address this issue, but require computational expertise to run and create a substantial barrier for researchers who are not comfortable with informatics.

The goal of this project is to address this barrier by building a well-tooled LLM-based agent capable of orchestrating the full cohort phenotyping pipeline end-to-end from a single natural-language research prompt. It extends the M4 natural-language SQL infrastructure [3] with a multimodal preprocessing and phenotyping layer, orchestrated through Claude Code skills that produce a reproducible, researcher-guided protocol from a single prompt instruction.

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2 Background (Brief)

EHR-based real-world evidence pipelines: Generating real-world evidence from EHRs requires a structured sequence of data curation steps. According to [1], a well-characterized integrated pipeline for this purpose encompasses four steps: (1) harmonizing raw codes to standardized vocabularies, (2) constructing a disease cohort via probabilistic phenotyping, (3) extracting relevant variables from structured and unstructured data, and (4) validating and robustly modeling outcomes, where each module draws on a distinct set of specialized methods and tools. Our work operationalizes modules one and two of this pipeline via an autonomous LLM agent, bridging the gap between raw EHR data and validated patient cohorts.

EHR phenotyping algorithms: There are many phenotyping algorithms in the literature, with different purposes and levels of supervision. Our project focuses on two state of the art methods: MAP (Multimodal Automated Phenotyping) [4], which scores patients probabilistically using co-occurrence of structured codes and NLP signal, and LATTE [5], which provides incident phenotyping in a label efficient way. For feature selection, ONCE (Online Narrative and Codified feature search Engine) [2] is a multi-source knowledge graph representation learning driven tool that has been proven to work slightly better than LLMs for feature selection.

AI agents for clinical data workflows: The emergence of tool-augmented LLM agents has fundamentally changed how a researcher can interact with data. The M4 project [3] demonstrated that LLM agents can reliably generate, execute, and self-correct SQL queries against MIMIC-IV, effectively solving the natural-language-to-SQL translation problem for clinical databases. More broadly, frameworks such as ToolUniverse [6] and AutoResearchClaw [7] have shown that equipping LLMs with domain-specific callable tools enables end-to-end research pipelines. However, none of these systems provide tools for vocabulary standardization, clinical NLP, or probabilistic cohort phenotyping. Our system targets this gap, and is designed as a module that could be integrated into these broader agentic clinical research frameworks once validated on MIMIC-IV.

3 Data

Our project uses MIMIC-IV as the primary EHR dataset to demonstrate proof of concept. MIMIC-IV is a large, deidentified dataset of patients admitted to the emergency department or intensive care unit at Beth Israel Deaconess Medical Center between 2008 and 2022, comprising over 65,000 ICU patients and more than 200,000 emergency department patients [8]. It is widely used in clinical research, publicly accessible, and uniquely valuable for this project because it includes both structured and unstructured data. Its companion dataset, MIMIC-IV-Note, provides deidentified free-text clinical notes, including over 331,794 discharge summaries [9]. While MIMIC-IV is used here for validation, our agentic pipeline is designed to be adaptable to any SQL-queryable EHR dataset. No manual inclusion/exclusion criteria or preprocessing steps are applied to MIMIC-IV, as these processes are intended to be handled autonomously by the agent.

4 Baseline Implementation (Required)

The baseline system is an LLM-based EHR query agent (built on M4 [3]) that constructs patient cohorts through code-based filtering on MIMIC-IV, without any structured preprocessing, NLP on unstructured notes, or probabilistic phenotyping. This represents the current state-of-the-art for natural-language clinical cohort building and serves as the comparison point for our pipeline. The baseline agent accepts a free-text research question, autonomously generates SQL queries against MIMIC-IV structured tables, and returns a cohort defined by the presence of one or more diagnosis codes matching the target condition. The agent must identify relevant codes from its parametric knowledge alone, with no access to multimodal preprocessing or advanced phenotyping tools. Claude Code was chosen as the orchestrating agent.

Our autonomous phenotyping system builds on this baseline by adding a suite of skills for structured code normalization, unstructured note processing, and probabilistic phenotyping. Rather than relying on code-based heuristic filtering, the agent uses clinically validated tools to standardize EHR vocabularies, extract signal from discharge notes, and score patients probabilistically using

state-of-the-art EHR phenotyping algorithms. The phenotyping algorithm that we use is MAP [4], a multimodal phenotyping algorithm that fits mixture models in aggregated ICD and NLP counts. MAP is an unsupervised multimodal phenotyping algorithm that estimates the probability that each patient’s feature co-occurrence pattern is consistent with true disease presence. For each patient, MAP returns a probability score and a binary label.

Both the baseline and our proposed pipelines are initiated from a single research prompt and execute reproducibly end-to-end within Claude Code. They return a set of Python scripts and Markdown files containing the research protocol, the query logic, the resulting patient cohort, a CONSORT-style flow diagram, summary statistics and other cohort characterizations, and an explanation of key findings and limitations.

4.1 Baseline Results

To contextualize baseline performance, we present preliminary results comparing the baseline agent against our proposed pipeline, which is described in full in Section 5. Because MIMIC-IV lacks ground-truth chart review labels, classical classification metrics such as AUROC are not computable. We instead evaluate pipeline performance through cohort composition analysis, treating the reduction in cohort size and the subset relationship between pipelines as proxies for improved specificity.

To evaluate the pipeline, we ran both agent configurations on multiple diseases in MIMIC-IV, including multiple sclerosis (MS) and rheumatoid arthritis (RA). We present RA as a detailed case study here. To check that our agent correctly ran MAP on RA, we expect to see a bimodal distribution of MAP scores between RA cases and controls, confirmed in Figure 1. The CONSORT-style flow diagrams generated by the baseline agent pipeline and our proposed agent pipeline are displayed in Figures 2 and 3, respectively. Notably, there is a substantial difference in the final generated RA cohort sizes between the baseline and our proposed pipeline: 3521 and 1398, respectively. To investigate this difference, we compared the patient IDs in both cohorts, illustrated in Figure 4 and further detailed in Table 1. We find that the proposed agent-produced cohort is almost a strict subset of the baseline agent-produced cohort, which provides promising results that our agent is using its phenotyping algorithm as it was designed: a multimodal high-precision filter that distinguishes true RA cases from rule-out diagnoses, miscodings, and billing artifacts. The phenotyping algorithm improves specificity over baseline without introducing new cases, keeping only around 40% of originally flagged patients—directionally consistent with the 58% statistic from the previously mentioned MGB cohort study [2]. Overall, these results demonstrate that our agentic pipeline successfully operationalizes probabilistic phenotyping end-to-end, correctly executing, interpreting, and applying it autonomously to produce high-precision cohorts.

5 Proposed Method: Design and Implementation Status

The proposed system is a multi-step agentic pipeline orchestrated via Claude Code that takes a single research prompt and autonomously executes cohort construction through sequential layers: EHR data access (handled through M4), structured data preprocessing and standardization, NLP extraction from discharge notes, feature selection, and probabilistic phenotyping.

For structured data, raw codes are normalized, currently across three vocabularies: ICD diagnosis codes are rolled up to PheCodes, CPT procedure codes are rolled up to CCS categories, and NDC medication codes are rolled up to RXNorm ingredient-level representations. This standardization collapses granular billing codes into clinically coherent groupings that serve as stable, interpretable features for downstream phenotyping. We plan on expanding these capabilities through the OMOP mappings in ATHENA (<https://athena.ohdsi.org/>), as well as giving the agent the knowledge necessary to write new mapping files for custom vocabularies.

For feature selection, this pipeline uses ONCE [2], a multisource knowledge graph based tool that identifies which codified concepts and which UMLS CUIs are most relevant to a condition of interest. The outputs of ONCE are two dictionaries, one for NLP extraction and one for codified feature selection. This is the only part of the pipeline that we kept manual, as the creators provide a webapp interface (<https://shiny.parse-health.org/ONCE/>) that works very well, and it allows the researcher to modify the provided dictionary, or adapt it to a specific usecase before starting the pipeline.

The NLP dictionary serves as extraction targets for MedSpaCy [10], a library of tools for clinical NLP, which scans discharge summaries for matching concept mentions. MedSpaCy addresses the unique linguistic challenges of clinical narratives, such as heavy use of abbreviations, negation, uncertain, family-history mentions and unstructured formatting, which often confound general-domain NLP tools. Although less accurate than a fine-tuned clinical LLM, this approach is substantially more computationally efficient and scalable given the large volume of notes that need to be processed, and produces sufficient signal for downstream phenotyping.

The structured and NLP features are jointly passed to a probabilistic phenotyping algorithm. Currently, we have implemented MAP [4], a multimodal phenotyping algorithm that fits mixture models on aggregated ICD and NLP counts. MAP is an unsupervised algorithm that estimates the probability that each patient’s feature co-occurrence pattern is consistent with true disease presence, returning a continuous posterior probability score between 0 and 1 and a binary case/control label per patient. MAP is natively implemented in R, so a Python subprocess bridge to the R-native MAP package was implemented so it can be called. We plan to integrate additional phenotyping algorithms—most notably an incident phenotyping algorithm called LATTE [5], as described in Section 5.1. The agent will have the knowledge necessary to decide which phenotyping algorithm, if any, to apply to the user’s research question. For example, if the user asks "Build a cohort of patients with rheumatoid arthritis", the agent will apply MAP to identify all patients who have had an RA diagnosis; if the user asks "Build a cohort of patients with rheumatoid arthritis and identify when they were first diagnosed", the agent will apply LATTE to pinpoint the date of each patient’s first RA diagnosis.

The pipeline is implemented as a suite of five agent skills. The `clinical-research-session` skill serves as the top-level orchestrator: it conducts a structured researcher interview, drafts a formal research protocol, and coordinates all downstream skill calls in a script-first workflow where every result-producing step is written to a stored Python script before execution. The `phenotyping-strategy` skill formalizes the decision logic for choosing between cohort filtering algorithms (rule-based versus probabilistic phenotyping), favoring probabilistic phenotyping when ONCE co-features are available, the candidate pool is large, or ICD specificity is known to be low. The `preprocessing` skill executes structured code normalization rollups and assembles all modalities into a unified long-format observation log. The `note-preprocessing` skill extracts CUI mentions from notes using MedSpaCy and appends them to the observation log. Finally, the `phenotyping` skill formats the input for the chosen phenotyping algorithm, invokes it, and returns a probability score and a binary phenotype label per patient.

5.1 Planned Experiments Beyond the Baseline

First, we will integrate LATTE [5], a temporal phenotyping algorithm that identifies the incident date at which a patient first develops a condition. This moves our pipeline from simple binary phenotyping towards a more clinically useful output that captures disease onset, enabling downstream analyses such as time-to-event modeling that are not possible with a prevalence cohort alone.

Second, we will perform a more rigorous evaluation of our pipeline than presented here. We plan to conduct a larger scale ablation experiment on multiple different ICU conditions, systematically evaluating the marginal contribution of each pipeline layer by running at least the following three conditions: code filtering only (the baseline agent); probabilistic phenotyping on structured features alone; and probabilistic phenotyping with full NLP. Comparing the first two conditions isolates the contribution of structured preprocessing and probabilistic phenotyping, while comparing the last two conditions isolates the added signal from unstructured notes. The resulting cohort sizes and phenotyping score distribution shapes will be compared for each condition, as well as the rate at which patients are reclassified after each subsequent layer is added.

Third, since MIMIC-IV does not have ground truth chart review, we may explore LLM-as-a-judge evaluation (time-permitting) as a scalable alternative to clinician chart review. Separate LLM instances would be provided a summary of each patient’s chart and asked to output a binary case/control judgment with a confidence rating. Agreement between judge confidence and phenotyping posterior probability would serve as evidence of the underlying phenotyping algorithms and our pipeline’s validity.

6 Challenges, Risks, and Mitigation

Technical challenges will arise with the planned integration of LATTE [5]: Because LATTE anchors phenotyping to an incident date, any features derived from after that date would constitute temporal leakage and corrupt the cohort definition. Ensuring the agent respects this boundary may require guardrails beyond prompt engineering alone. Our planned mitigation is to implement automated leakage detection checks that run after our pipeline executes, flagging any temporal violations before the cohort is used downstream.

Another practical challenge we anticipate facing is context rot: as tool calls, script outputs, and intermediate reasoning accumulate in our multi-step agent pipeline, the context window fills rapidly and reasoning quality degrades. In our MAP phenotyping runs, we already reach 50–75% of the Claude Code context limit within a single session, and adding LATTE to our pipeline will further increase this number. Our current mitigation is Claude Code’s native summarization (`/compact`), which condenses earlier conversation history into a summary. If that proves to be insufficient, we may restructure our pipeline as a multi-agent system, where a thin orchestrator delegates each skill invocation to a dedicated sub-agent that executes, writes its outputs to files, and terminates, keeping the orchestrator context clean and minimal.

7 Updated Timeline and Plan to Final Report

The plan for the final report is:

- Week 9: Implement LATTE, update phenotyping skills, and solve temporality-related issues.
- Week 10: Test both algorithms on more conditions; finalize preprocessing tools.
- Week 11: Implement LLM-as-a-judge.
- Week 12: Buffer week to solve challenges, wrap project repo in a deliverable way.
- Week 13: Conduct remaining analyses, make slides and figures, and write final report.

8 Figures

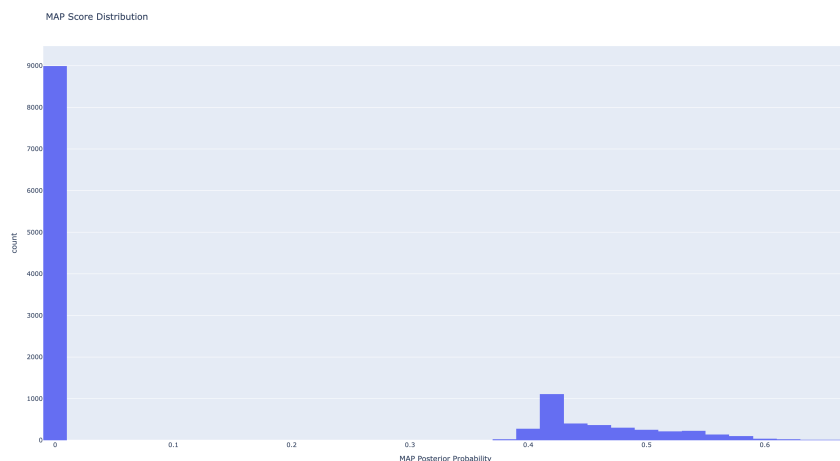


Figure 1: Histogram of MAP scores for RA produced from our agentic pipeline, with the MAP posterior probability on the x-axis and the number of patients on the y-axis.

Table 1: Final RA cohort sizes produced by baseline and MAP agent pipelines.

Agent Pipeline	Cohort Size
MAP	1,398
Baseline	3,521
Shared	1,395
Only in MAP	3
Only in Baseline	2,126

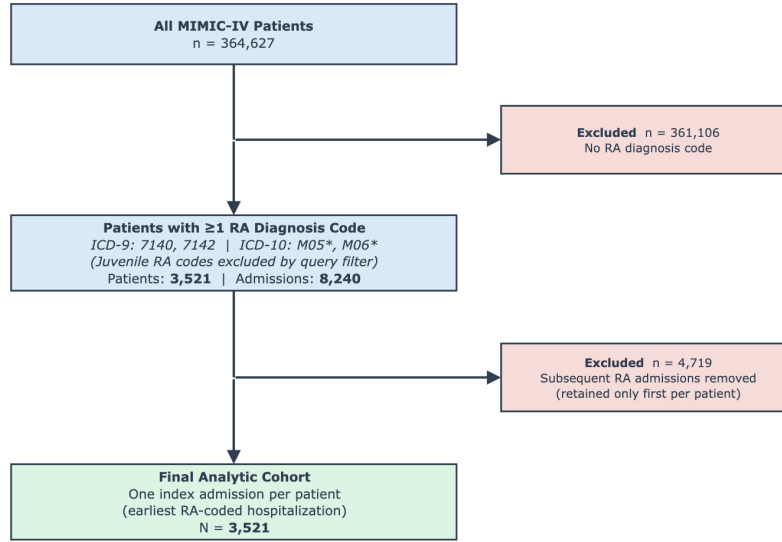


Figure 2: CONSORT-style flow diagram of RA cohort generated by baseline agent pipeline.

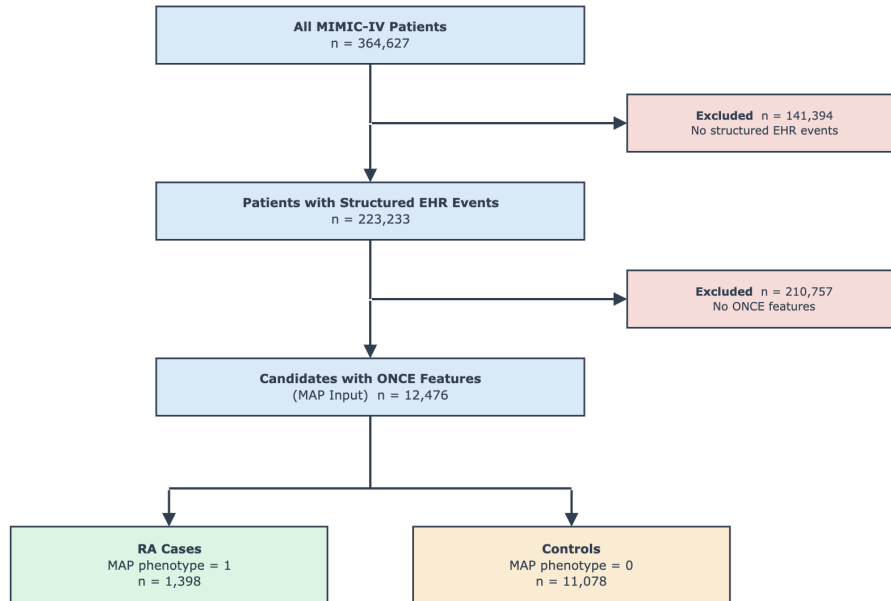


Figure 3: CONSORT-style flow diagram of RA cohort generated by MAP agent pipeline.

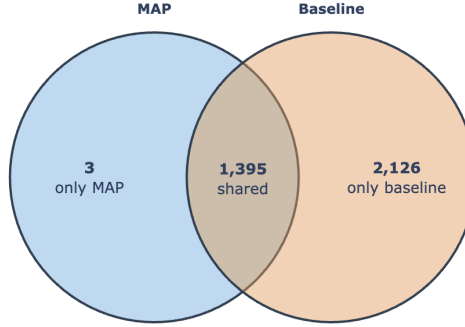


Figure 4: Venn diagram of RA cohorts produced by baseline and MAP agent pipelines (visualizing the same information displayed in Table 1).

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